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Journal of Approximation Theory 179 (2014) 112–127

JOURNAL OF Approximation Theory

www.elsevier.com/locate/jat

Full length article

Orthogonal matrix polynomials whose differences are also orthogonal☆

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Received 10 July 2013; received in revised form 19 November 2013; accepted 24 November 2013
Available online 12 December 2013

Communicated by Guillermo López Lagomasino

Abstract

We characterize orthogonal matrix polynomials $(P_n)_n$ whose differences $(\nabla P_{n+1})_n$ are also orthogonal by means of a discrete Pearson equation for the weight matrix W with respect to which the polynomials $(P_n)_n$ are orthogonal. We also construct some illustrative examples. In particular, we show that contrary to what happens in the scalar case, in the matrix orthogonality the discrete Pearson equation for the weight matrix W is, in general, independent of whether the orthogonal polynomials with respect to W are eigenfunctions of a second order difference operator with polynomial coefficients.
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Keywords: Orthogonal matrix polynomials; Difference equations; Difference operators; Charlier polynomials; Matrix orthogonality

1. Introduction

Classical discrete orthogonal polynomials (Charlier, Meixner, Krawtchouk and Hahn) are characterized by a number of equivalent properties. The following ones are three of them:

1. They are eigenfunctions of a second order difference operator of the form

$$\sigma \Delta\nabla + \tau \Delta,$$

☆ Partially supported by MTM2012-36732-C03-03 (Ministerio de Economía y Competitividad), FQM-262, FQM-4643, FQM-7276 (Junta de Andalucía) and Feder Funds (European Union).
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http://dx.doi.org/10.1016/j.jat.2013.11.012

where σ and τ are polynomials of degrees not bigger than 2 and 1, respectively, and Δ and ∇ denote the usual first order difference operators $\Delta(p) = p(x+1) - p(x)$ and $\nabla(p) = p(x) - p(x-1)$.

2. The sequence of its differences is again orthogonal with respect to a measure.
3. Each classical discrete weight measure w satisfies a discrete Pearson difference equation $\Delta(g_2 w) = g_1 w$, where g_1 and g_2 are polynomials of degrees at most 1 and 2, respectively.

The purpose of this paper is to show that for orthogonal matrix polynomials these three properties are no longer equivalent. More precisely, we prove that the equivalence between (2) and (3) above still remains true for orthogonal matrix polynomials. Under suitable Hermitian assumptions on wg_2 and wg_1, they also imply the property (1), but the converse is not true.

A matrix moment functional is a (left) linear functional $u : \mathbb{P}^N \to \mathbb{C}^{N\times N}$, where $\mathbb{P}^N$ denotes the linear space of $N \times N$ matrix polynomials. In this paper, we consider orthogonality with respect to a matrix moment functional, that is, with respect to the bilinear form $\langle \cdot, \cdot \rangle_u : \mathbb{P}^N \times \mathbb{P}^N \to \mathbb{C}^{N\times N}$ defined by $\langle P, Q\rangle_u = \sum_{i,j} A_i u(x^{i+j} I) B_j^*$, where $P = \sum_n A_n x^n$ and $Q = \sum_n B_n x^n$. We then say that a sequence of matrix polynomials $(P_n)_n$, P_n of degree n with nonsingular leading coefficient, is orthogonal with respect to u if $\langle P_n, P_k\rangle_u = \Lambda_n \delta_{k,n}$, where Λ_n is a nonsingular matrix for $n \geq 0$. We say that the moment functional u is quasi-definite if it has a sequence of orthogonal polynomials. The most interesting case appears when the moment functional is defined by a weight matrix W. A weight matrix W is a $N \times N$ matrix of measures supported in the real line such that $W(A)$ is positive semidefinite for any Borel set $A \subset \mathbb{R}$, having finite moments and satisfying that $\int P(x) dW(x) P^*(x)$ is nonsingular if the leading coefficient of the matrix polynomial P is nonsingular. Each weight matrix W defines a matrix moment functional, which we denote also by W: $W(P) = \int P dW$. The corresponding bilinear form is then defined by $\langle P, Q\rangle_W = \int P dW Q^*$. The theory of matrix valued orthogonal polynomials starts with two papers by M. G. Kreĭn in 1949, see [9,10].

The paper is organized as follows. Section 2 will be devoted to basic definitions and facts. In particular, we define the operational calculus with moment functionals. As usual, the basic operations such as multiplication by polynomials or the first order difference operators will be defined by duality. In particular, for a polynomial $P = \sum_n A_n x^n$ and a moment functional u, we define the moment functionals Pu and uP as

$$Pu(Q) = u(QP), \tag{1.1}$$

$$uP(Q) = \sum_n u(Qx^n) A_n. \tag{1.2}$$

Notice that for a polynomial $p \in \mathbb{P}$, we have $pIu = upI$. With these definitions, one straightforwardly has $\langle P, Q\rangle_u = \langle P, uQ^*\rangle$. These definitions agree with the usual product of a polynomial P times a weight matrix W. For a moment functional u, the new moment functionals Δu and ∇u are defined by

$$\Delta u(P) = -u(\nabla P), \qquad \nabla u(P) = -u(\Delta P), \tag{1.3}$$

respectively.

In Section 3, we prove the equivalence between the properties (2) and (3) above for matrix moment functionals. More precisely, we prove the following theorem.

Theorem 1. *Let u be a quasi-definite moment functional. Consider a sequence of orthogonal matrix polynomials $(P_n)_n$ with respect to u. Then, the following conditions are equivalent:*

(1) *The functional u satisfies the discrete Pearson equation $\Delta(uG_2^*) = uG_1^*$, where $G_2 = G_{2,2}x^2 + G_{2,1}x + G_{2,0}$ and $G_1 = G_{1,1}x + G_{1,0}$ are matrix polynomials of degrees not bigger than 2 and 1, respectively, satisfying that $(n-1)G_{22} + G_{11}$ is not singular for $n \geq 1$.*
(2) *The sequence $(\nabla P_{n+1})_n$ is a family of orthogonal matrix polynomials with respect to a quasi-definite moment functional $\widetilde{u}$.*

Furthermore, $\widetilde{u} = uG_2^*$.

In Section 4, we prove that, under additional assumptions, the discrete Pearson equation $\Delta(uG_2^*) = uG_1^*$ implies that the orthogonal polynomials $(P_n)_n$ with respect to u are eigenfunctions of a second order difference operator.

Theorem 2. *Let u be a quasi-definite moment functional. Consider a sequence of orthogonal matrix polynomials $(P_n)_n$ with respect to u. Assume that the moment functional u satisfies the discrete Pearson equation $\Delta(uG_2^*) = uG_1^*$, and the Hermitian conditions $G_2u = uG_2^*$ and $G_1u = uG_1^*$, where G_2 and G_1 are polynomials of degrees at most 2 and 1, respectively. Consider the second order difference operator*

$$D = \Delta\nabla(\cdot)G_2 + \Delta(\cdot)G_1.$$

Then $D(P_n) = \Lambda_n P_n$, for certain matrices Λ_n.

For the differential version of these theorems see [2].

In Section 5 we display some illustrative examples. For a positive real number $a > 0$, and non null complex numbers v_i, $i = 1, \ldots, N-1$, we first consider the family of $N \times N$ weight matrices

$$W = \sum_{x=0}^{\infty} \frac{a^x}{x!}(I + A)^x (I + A^*)^x \delta_x, \tag{1.4}$$

where A is the nilpotent matrix (of order N)

$$A = \begin{pmatrix} 0 & v_1 & 0 & \cdots & 0 \\ 0 & 0 & v_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & v_{N-1} \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}. \tag{1.5}$$

These weight matrices were introduced in [1] where it is proved that the orthogonal polynomials with respect to W are also eigenfunctions of a second order difference operator of the form

$$\Delta\nabla(\cdot)F_2 + \Delta(\cdot)F_1 + \cdot F_0. \tag{1.6}$$

In this paper, we assume that the parameters v_i, $i = 1, \ldots, N-1$, satisfy the constraints

$$(N-i-1)a|v_i|^2|v_{N-1}|^2 + (N-1)|v_i|^2 - i(N-i)|v_{N-1}|^2 = 0. \tag{1.7}$$

Denote by $\mathcal{A}$ the set formed by all nilpotent matrices of the form (1.5) whose entries v_i, $i = 1, \ldots, N-1$, satisfy the constraints (1.7).

We then construct matrix polynomials G_2 and G_1 of degrees 2 and 1, respectively, such that $\Delta(WG_2^*) = WG_1^*$, and $G_2W = WG_2^*$. As a consequence of Theorem 1, we get that the orthogonal polynomials $(P_n)_n$ with respect to W have differences $(\nabla P_{n+1})_{n\geq 0}$ which are also

orthogonal. The orthogonality weight matrix for $(\nabla P_{n+1})_{n\geq 0}$ is $\widetilde{W} = G_2 W$ which it turns to be a weight matrix again (supported in the set of positive integer $\{1, 2, \ldots\}$). Moreover, we prove that $\widetilde{W}$ is essentially of the form (1.4): indeed, there are a nonsingular matrix M and a nilpotent matrix $\widetilde{A} \in \mathcal{A}$ such that $\widetilde{W}(x) = M W_{\widetilde{A}}(x-1) M^*$ for all $x \geq 1$. In particular, this implies that for all $k \geq 1$, the differences of order k, $(\nabla^k P_{n+k})_{n\geq 0}$ are again orthogonal polynomials.

Theorem 2 also implies that the orthogonal polynomials $(P_n)_n$ are eigenfunctions of the second order difference operator

$$D = \Delta\nabla(\cdot) G_2 + \Delta(\cdot) G_1.$$

This difference operator is however linearly independent to the one found in [1]. This shows once again one of the striking developments in this area: the discovery of many new phenomena that are absent in the well known scalar theory. One of these phenomena is that the elements of a family of orthogonal matrix polynomials can be eigenfunctions of several linearly independent second order difference operators while the elements of each of the classical discrete families in the scalar case are eigenfunctions of only one linearly independent second order difference operator, respectively (see [5] for more details; for the differential version of this result see [4,6] and the references therein).

We study in more detail the case 3×3, where we consider the weight matrix W defined by (1.4) where A is now an arbitrary 3×3 upper triangular nilpotent matrix

$$A = \begin{pmatrix} 0 & v_1 & v_3 \\ 0 & 0 & v_2 \\ 0 & 0 & 0 \end{pmatrix}.$$

We show that there are always polynomials G_2 and G_1 of degrees not bigger than 2 and 1, respectively, such that $\Delta(W G_2^*) = W G_1^*$. However, only when $v_3 = 0$ and the parameters v_1 and v_2 satisfy (1.7), one of these polynomials $G_2 \neq 0$ satisfies also the Hermitian condition $G_2 W = W G_2^*$. This means that the polynomials $(\nabla P_{n+1})_{n\geq 0}$ are always orthogonal with respect to a moment functional but this moment functional can be represented by a weight matrix (in the sense defined above) only when $v_3 = 0$ and the parameters v_1 and v_2 satisfy (1.7). We also show that, in general, the orthogonal polynomials $(P_n)_n$ are not eigenfunctions of a second order difference operator (unless $v_3 = 0$). This shows that the hypothesis $G_2 W = W G_2^*$ cannot be removed in Theorem 2.

We finally display a weight matrix W whose orthogonal polynomials $(P_n)_n$ are eigenfunctions of a second order difference operator of the form (1.6). However this weight matrix W does not satisfy any discrete Pearson equation. This shows that in the matrix orthogonality, the discrete Pearson equation for the weight matrix W is independent of whether the orthogonal polynomials with respect to W are eigenfunctions of a second order difference operator of the form (1.6).

2. Preliminaries

We will denote by $\mathbb{C}^{N\times N}$ the set of matrices of size $N \times N$ whose elements are complex numbers, and by $\mathbb{P}^N$ the linear space of matrix polynomials in one real variable with coefficients in $\mathbb{C}^{N\times N}$. The linear subspace of $\mathbb{P}^N$ formed by those matrix polynomials with degree at most n will be denoted by $\mathbb{P}_n^N$. $\mathbb{P}$ will stand for the linear space of polynomials with complex coefficients.

A moment functional u is a left linear functional $u : \mathbb{P}^N \to \mathbb{C}^{N\times N}$: $u(AP + BQ) = Au(P) + Bu(Q)$, with $A, B \in \mathbb{C}^{N\times N}$ and $P, Q \in \mathbb{P}^N$. For a moment functional u and a matrix polynomial $P \in \mathbb{P}^N$, the duality bracket is defined by $\langle P, u\rangle = u(P)$. We will use the

duality bracket $\langle P, u\rangle$ instead of the functional notation $u(P)$. Notice that we write the moment functional in the right hand side of the bracket contrarily to the standard notation. This is because of the following reason. As we will explain below a moment functional can always be represented by integrating with respect to a matrix of measures W: $\langle P, u\rangle = \int P(t)dW(t)$. We write the moment functional in the right hand side of the bracket to stress that in this representation the matrix of measures is multiplying on the right.

We associate to each moment functional u a bilinear form $\langle\cdot,\cdot\rangle_u : \mathbb{P}^N \times \mathbb{P}^N \to \mathbb{C}^{N\times N}$ in the linear space of matrix polynomials defined by

$$\langle P, Q\rangle_u = \sum_{i,j} A_i \langle x^{i+j} I, u\rangle B_j^*, \tag{2.1}$$

where $P = \sum_n A_n x^n$ and $Q = \sum_n B_n x^n$. We then say that a sequence of matrix polynomials $(P_n)_n$, P_n of degree n with nonsingular leading coefficient, is orthogonal with respect to the moment functional u if $\langle P_n, P_k\rangle_u = \Lambda_n \delta_{k,n}$, where Λ_n is a nonsingular matrix for $n \geq 0$. An easy consequence of the orthogonality is that orthogonal polynomials $(P_n)_n$ with respect to a moment functional satisfy a three term recurrence relation

$$x P_n(x) = \alpha_n P_{n+1}(x) + \beta_n P_n(x) + \gamma_n P_{n-1}(x), \quad n \geq 0, \tag{2.2}$$

where $P_{-1} = 0$, and $\alpha_n, \gamma_n, n \geq 1$, are non-singular matrices.

We say that a moment functional u is quasi-definite if $\det(\Delta_n) \neq 0$ for every $n \geq 0$, where Δ_n is the Hankel-block matrix

$$\Delta_n = \begin{pmatrix} \mu_0 & \mu_1 & \cdots & \mu_n \\ \mu_1 & \mu_2 & \cdots & \mu_{n+1} \\ \vdots & \vdots & & \vdots \\ \mu_n & \mu_{n+1} & \cdots & \mu_{2n} \end{pmatrix}$$

and μ_k is the moment of order k with respect to u: $\mu_k = \langle x^k I, u\rangle$. Notice that the moments μ_k, $k \geq 0$, of u define uniquely both the moment functional u and the bilinear form $\langle\cdot,\cdot\rangle_u$ (2.1). It turns out that a moment functional u has a sequence of orthogonal polynomials if and only if u is quasi-definite (see [2]).

Given a moment functional u with moments μ_k, $k \geq 0$, we define its (complex) adjoint u^* as the moment functional with moments μ_k^*, $k \geq 0$. If the moments μ_k, $k \geq 0$, of a moment functional u are Hermitian matrices, i.e. $u = u^*$, we say that u is Hermitian. If, in addition, $\det(\Delta_n) > 0$ for every $n \geq 0$, we say that u is positive definite.

Moment functionals can always be represented by integration with respect to a matrix of measures: using [7], one can see that given a moment functional u there exists a matrix of measures W supported in the real line (whose entries are Borel measures in $\mathbb{R}$) such that $\langle P, u\rangle = \int_{\mathbb{R}} P dW$. The most interesting case is when the matrix moment functional is defined by a weight matrix W. A weight matrix W is an $N \times N$ matrix of measures supported in the real line such that $W(A)$ is positive semidefinite for any Borel set $A \subset \mathbb{R}$, having finite moments and satisfying that $\int P(x)dW(x)P^*(x)$ is nonsingular if the leading coefficient of the matrix polynomial P is nonsingular. It follows from [8] that a moment functional can be represented by a weight matrix if and only if it is positive definite.

Along this paper, we will use without an explicit mention the usual properties listed below of the difference operators Δ and ∇

1. $\nabla\Delta = \Delta\nabla = \Delta - \nabla$.
2. $\Delta f(x) = \nabla f(x+1)$.
3. $\Delta[f(x)g(x)] = f(x)\Delta g(x) + [\Delta f(x)]\, g(x+1)$,
4. $\nabla[f(x)g(x)] = f(x-1)\nabla g(x) + [\nabla f(x)]\, g(x)$.
5. $\sum_a^{b-1} f(x)\Delta g(x) = f(x)g(x)\Big|_a^b - \sum_a^{b-1} [\Delta f(x)]\, g(x+1)$,
6. $\sum_{a+1}^{b} f(x)\nabla g(x) = f(x)g(x)\Big|_a^b - \sum_{a+1}^{b} [\nabla f(x)]\, g(x-1)$.

The difference operators Δ and ∇ acting on moment functional are defined by

$$\langle P, \Delta u\rangle = -\langle \nabla(P), u\rangle, \qquad \langle P, \nabla u\rangle = -\langle \Delta(P), u\rangle. \tag{2.3}$$

A simple computation gives that

$$\begin{aligned} \Delta(Pu) &= P(x+1)\Delta u + (\Delta P)u, & \nabla(Pu) &= P(x-1)\nabla u + (\nabla P)u,\\ \Delta(uP) &= (\Delta u)P(x+1) + u\Delta P, & \nabla(uP) &= (\nabla u)P(x-1) + u\nabla P. \end{aligned}$$

Given a discrete weight matrix

$$W = \sum_{x=a}^{b} W(x)\delta_x,$$

supported in $\{a, a+1, \ldots, b-1, b\}$ (a can be $-\infty$ and b can be $+\infty$), the matrices of measures ΔW and ∇W are defined in the usual way by

$$\Delta W = \sum_{x=a-1}^{b} (W(x+1) - W(x))\delta_x, \tag{2.4}$$

$$\nabla W = \sum_{x=a}^{b+1} (W(x) - W(x-1))\delta_x, \tag{2.5}$$

respectively, where by definition $W(b+1) = W(a-1) = 0$ (if $a = -\infty$ or $b = \infty$, we take $a - 1 = -\infty$ or $b + 1 = +\infty$, respectively). It is worth noting that the support of ΔW is $\{a-1, a, a+1, \ldots, b-1, b\}$ and that it is different to the support of W, except when $a = -\infty$. In the same way, the support of ∇W is $\{a, a+1, \ldots, b-1, b, b+1\}$ and it is different to the support of W, except when $b = +\infty$. A simple computation shows that if we consider the weight matrix W as a moment functional these definitions of ΔW and ∇W agree with (2.3).

We finish this section introducing the concept of quasi-orthogonal polynomials with respect to a moment functional u. A family of polynomials $(P_n)_n$, P_n of degree n with nonsingular leading coefficient, is quasi-orthogonal of order r with respect to a moment functional u if $\langle P_n, x^k I\rangle_u = 0$ for $k = 0, \ldots, n-r-1$.

Theorem 3.4 of [3] shows that if $(P_n)_n$ are orthogonal with respect to the moment functional u and quasi-orthogonal of order r with respect to the moment functional $\widetilde{u}$ then there exists a matrix polynomial Q of degree r such that $\widetilde{u} = uQ$.

3. Orthogonal polynomials whose differences are also orthogonal

In this section we prove Theorem 1 in the Introduction.

Proof. (1) ⇒ (2). A simple computation using the definitions and basic properties introduced in the previous section gives

$$\begin{aligned}\langle \nabla P_n, x^k I\rangle_{uG_2^*} &= \langle \nabla P_n, uG_2^* x^k\rangle = -\langle P_n, \Delta(uG_2^* x^k)\rangle \\ &= -\langle P_n, \Delta(uG_2^*)(x+1)^k + uG_2^*\Delta(x^k)\rangle \\ &= -\langle P_n, (x+1)^k G_1\rangle_u - \langle P_n, \Delta(x^k)G_2\rangle_u.\end{aligned}$$

If $0 \le k \le n-2$, since $(x+1)^k G_1$ and $\Delta(x^k)G_2$ are polynomials of degree at most $n-1$, we get

$$\langle \nabla P_n, x^k I\rangle_{uG_2^*} = 0. \tag{3.1}$$

For $k = n-1$, we have

$$\begin{aligned}\langle \nabla P_n, x^{n-1} I\rangle_{uG_2^*} &= -\langle P_n, (x+1)^{n-1} G_1\rangle_u - \langle P_n, \Delta(x^{n-1})G_2\rangle_u \\ &= -\langle P_n, x^n I\rangle_u G_{11}^* - (n-1)\langle P_n, x^n I\rangle_u G_{22}^* \\ &= -\langle P_n, x^n I\rangle_u [G_{11} + (n-1)G_{22}]^*. \end{aligned} \tag{3.2}$$

Since both matrices $\langle P_n, x^n I\rangle_u$ and $G_{11}^* + (n-1)G_{22}^*$ are nonsingular matrices, we get that also $\langle \nabla P_n, x^{n-1} I\rangle_{uG_2^*}$ is a nonsingular matrix.

This proves the orthogonality of the family $(\nabla P_{n+1})_n$ with respect to uG_2^*.

(2) ⇒ (1). We start writing P_n as a linear combination of ∇P_{n-1}, ∇P_n and ∇P_{n+1}.

Since $(P_n)_n$ is a family of orthogonal polynomials, they satisfy a three term recurrence relation

$$x P_n = \alpha_n P_{n+1} + \beta_n P_n + \gamma_n P_{n-1}, \quad n \ge 0,$$

where we define $P_{-1} = 0$.

Applying now the operator ∇, we get

$$(\nabla x) P_n + (x-1)\nabla P_n = \alpha_n \nabla P_{n+1} + \beta_n \nabla P_n + \gamma_n \nabla P_{n-1},$$

and thus

$$P_n = \alpha_n \nabla P_{n+1} - x\nabla P_n + (\beta_n + 1)\nabla P_n + \gamma_n \nabla P_{n-1}. \tag{3.3}$$

On the other hand, since $(\nabla P_{n+1})_n$ is also a family of orthogonal polynomials, they satisfy a three term recurrence relation

$$x\nabla P_n = \widetilde{\alpha}_n \nabla P_{n+1} + \widetilde{\beta}_n \nabla P_n + \widetilde{\gamma}_n \nabla P_{n-1}, \quad n \ge 1.$$

Replacing in (3.3) we obtain

$$\begin{aligned} P_n &= \alpha_n \nabla P_{n+1} - \widetilde{\alpha}_n \nabla P_{n+1} - \widetilde{\beta}_n \nabla P_n - \widetilde{\gamma}_n \nabla P_{n-1} + (\beta_n + 1)\nabla P_n + \gamma_n \nabla P_{n-1} \\ &= a_n \nabla P_{n+1} + b_n \nabla P_n + c_n \nabla P_{n-1}, \end{aligned} \tag{3.4}$$

where $a_n = \alpha_n - \widetilde{\alpha}_n$, $b_n = \beta_n + 1 - \widetilde{\beta}_n$ and $c_n = \gamma_n - \widetilde{\gamma}_n$.

We now prove that $(P_n)_n$ is quasi-orthogonal of order 2 with respect to $\widetilde{u}$, i.e., $\langle P_n, x^k I\rangle_{\widetilde{u}} = 0$ for $k = 0, \dots, n-3$.

Since $(\nabla P_{n+1})_n$ is orthogonal with respect to $\widetilde{u}$, it follows that $\langle \nabla P_n, x^k I\rangle_{\widetilde{u}} = 0$ for $k \le n-2$, thus

$$\langle P_n, x^k I\rangle_{\widetilde{u}} = \left\langle a_n \nabla P_{n+1} + b_n \nabla P_n + c_n \nabla P_{n-1}, x^k I\right\rangle_{\widetilde{u}} = 0 \tag{3.5}$$

for k from 0 to $n-3$. Theorem 3.4 of [3] implies that there exists a matrix polynomial G_2^* of degree 2 such that $\widetilde{u} = uG_2^*$.

We next show that $(P_n)_n$ are quasi orthogonal of order 1 with respect to $\Delta(uG_2^*)$. Indeed

$$\begin{aligned}\langle P_n, x^k I\rangle_{\Delta(uG_2^*)} &= \langle x^k P_n, I\rangle_{\Delta(uG_2^*)} = \langle x^k P_n, \Delta(uG_2^*)\rangle = -\langle \nabla(x^k P_n), uG_2^*\rangle \\ &= -\langle (x-1)^k \nabla P_n, uG_2^*\rangle - \langle \nabla(x^k) P_n, uG_2^*\rangle \\ &= -\langle \nabla P_n, uG_2^*(x-1)^k\rangle - \langle P_n, uG_2^*\nabla(x^k)\rangle \\ &= -\langle \nabla P_n, (x-1)^k I\rangle_{uG_2^*} - \langle P_n, \nabla(x^k) I\rangle_{uG_2^*}.\end{aligned}$$

The orthogonality of ∇P_{n+1} with respect to uG_2^* and (3.5) imply that $\langle P_n, x^k I\rangle_{\Delta(uG_2^*)} = 0$ for $0 \le k \le n-2$. This proves that $(P_n)_n$ is a quasi-orthogonal family of order 1 with respect to $\Delta(uG_2^*)$, and according to Theorem 3.4 of [3] there exists a matrix polynomial G_1^* of degree 1 such that $\Delta(uG_2^*) = uG_1^*$.

We finally prove that the matrices $G_{11} + (n-1)G_{22}$ are nonsingular for $n \ge 1$. Indeed, proceeding as in the proof of (1) $\Rightarrow$ (2), we get (see (3.2))

$$\langle \nabla P_n, x^{n-1} I\rangle_{uG_2^*} = -\langle P_n, x^n I\rangle_u [G_{11} + (n-1)G_{22}]^*.$$

Since the matrix $\langle \nabla P_n, x^{n-1} I\rangle_{uG_2^*}$ is nonsingular for $n \ge 1$, we conclude that also $G_{11} + (n-1)G_{22}$ is nonsingular. □

It is easy to see that, under the additional assumption $uG_2^* = G_2 u$, the condition

$$\langle \nabla P_n, x^k I\rangle_{uG_2^*} = 0, \quad k \le n-2$$

(see (3.1)) is equivalent to

$$\nabla P_n G_2 = f_n P_{n+1} + g_n P_n + h_n P_{n-1},$$

for certain matrices f_n, g_n and h_n. In the scalar orthogonality, this last identity is usually called the first structure relation. The identity

$$P_n = a_n \nabla P_{n+1} + b_n \nabla P_n + c_n \nabla P_{n-1}$$

(see (3.4)) is called the second structure relation. Hence, in the previous proof we have also shown the equivalence between the conditions of Theorem 1 and the first and second structure relations (under the additional assumption $uG_2^* = G_2 u$).

4. Discrete Pearson equations and second order difference operators

In this section we prove Theorem 2 in the Introduction.

To do that, we introduce the concept of symmetry. For a linear operator $D : \mathbb{P}^N \to \mathbb{P}^N$, we say that D is symmetric with respect to the moment functional u if $\langle D(P), Q\rangle_u = \langle P, D(Q)\rangle_u$, for all polynomials $P, Q \in \mathbb{P}^N$.

Lemma 4.1. *Let D be a symmetric operator with respect to the quasi-definite moment functional u. Assume that in addition we have $\deg(D(P)) \le \deg(P)$ for every polynomial $P \in \mathbb{P}^N$. Then, the orthogonal polynomials with respect to u are also eigenfunctions of D.*

Proof. Indeed, one can write $D(P_n) = \sum_{k=0}^{n} A_k P_k$. Then for $k = 0, \ldots, n-1$, we have

$$A_k \langle P_k, P_k \rangle_u = \langle D(P_n), P_k \rangle_u = \langle P_n, D(P_k) \rangle_u = 0,$$

(since $D(P_k)$ is a polynomial of degree less than n) and thus $A_k = 0$. Hence $D(P_n) = A_n P_n$. □

We are now ready to prove Theorem 2.

Proof. Using the previous lemma, it is enough to prove that $\deg(D(P)) \leq \deg(P)$ for all polynomials $P \in \mathbb{P}$, and that the operator D is symmetric with respect to the moment functional u. The first condition follows easily taking into account that $\deg(G_2) \leq 2$ and $\deg(G_1) \leq 1$. The symmetry of D with respect to u is equivalent to prove that $\langle D(x^i I), x^j I \rangle_u = \langle x^i I, D(x^j I) \rangle_u$, $i, j \geq 0$.

Taking into account that $G_2 u = u G_2^*$ and $G_1 u = u G_1^*$, the Pearson equation can be rewritten as $\Delta(G_2 u) = G_1 u$. It is easy to see that this equation is equivalent to the equation $\nabla(G_1 u + G_2 u) = G_1 u$. Using it, the proof now follows from the following computations (where we use the definitions and basic properties of first order difference operators which can be found in Section 2).

$$\begin{aligned}
\langle D(x^i I), x^j I \rangle_u &= \langle D(x^i I), u x^j I \rangle = \langle ((\Delta \nabla x^i) G_2 + (\Delta x^i) G_1, u x^j \rangle \\
&= \langle \Delta \nabla x^i I, G_2 u x^j \rangle + \langle \Delta x^i I, G_1 u x^j \rangle \\
&= \langle x^i I, \nabla \Delta (G_2 u x^j) \rangle - \langle x^i I, \nabla (G_1 u x^j) \rangle \\
&= \langle x^i I, \nabla (\Delta (G_2 u)(x+1)^j + G_2 u \Delta x^j - G_1 u x^j) \rangle \\
&= \langle x^i I, \nabla ((G_2 u + G_1 u) \Delta x^j) \rangle \\
&= \langle x^i I, (\nabla (G_2 u + G_1 u)) \Delta (x-1)^j + (G_2 u + G_1 u) \nabla \Delta x^j \rangle \\
&= \langle x^i I, G_1 u \Delta (x-1)^j + G_1 u (\Delta x^j - \Delta (x-1)^j) + G_2 u \nabla \Delta x^j \rangle \\
&= \langle x^i I, G_2 u \nabla \Delta x^j + G_1 u \Delta x^j \rangle \\
&= \langle x^i I, u G_2^* \nabla \Delta x^j + u G_1^* \Delta x^j \rangle \\
&= \langle x^i I, (\nabla \Delta x^j) G_2 + (\Delta x^j) G_1 \rangle_u = \langle x^i I, D(x^j I) \rangle_u. \quad \square
\end{aligned}$$

5. Examples

In this section we consider three illustrative examples. Examples 1 and 2 illustrate Theorems 1 and 2 (in particular, that the assumptions $G_2 W = W G_2^*$ and $G_1 W = W G_1^*$ are necessary in Theorem 2). Example 3 shows that in the matrix orthogonality, the discrete Pearson equation for the weight matrix W does not imply that the orthogonal polynomials with respect to W are eigenfunctions of a second order difference operator of the form (1.6).

Example 1. For a positive real number $a > 0$, and non null complex numbers v_i, $i = 1, \ldots, N-1$, we consider the family of $N \times N$ weight matrices

$$W = \sum_{x=0}^{\infty} \frac{a^x}{x!} (I + A)^x (I + A^*)^x \delta_x, \tag{5.1}$$

where A is the nilpotent matrix defined by (1.5).

Consider also the diagonal matrix

$$J = \begin{pmatrix} N-1 & 0 & \cdots & 0 & 0 \\ 0 & N-2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix},$$

and the weight matrix $\widetilde{W}$ defined by

$$\widetilde{W} = \sum_{x=1}^{\infty} \frac{a^x}{(x-1)!} (I+A)^x \left(\frac{N-1}{a|v_{N-1}|^2} I + J \right) (I+A^*)^x \delta_x. \tag{5.2}$$

The following theorem provides an example with arbitrary size $N \times N$ of a weight matrix whose orthogonal polynomials have differences which are also orthogonal with respect to a weight matrix.

Theorem 3. *Let W and $\widetilde{W}$ be the weight matrices defined by (5.1) and (5.2), where we assume that the parameters v_i, $i = 1, \ldots, N-2$, satisfy the constraints*

$$(N-i-1)a|v_i|^2|v_{N-1}|^2 + (N-1)|v_i|^2 - i(N-i)|v_{N-1}|^2 = 0. \tag{5.3}$$

Let G_2 and G_1 be the matrix polynomials of degrees 2 and 1, respectively, given by

$$G_2(x) = \left[(I+A)^{-1} - I \right] x^2 + \left(\frac{N-1}{a|v_{N-1}|^2} I + J \right) x, \tag{5.4}$$

$$\begin{aligned} G_1(x) \; = \; & \left[J - \left(N + \frac{N-1}{a|v_{N-1}|^2} \right) I - aA + (I+A)^{-1} \right] x \\ & + a(I+A) \left(\frac{N-1}{a|v_{N-1}|^2} I + J \right) (I+A^*). \end{aligned} \tag{5.5}$$

Then, we have

1. $\widetilde{W} = G_2 W$, *and hence* $G_2 W = W G_2^*$.
2. $\Delta(G_2 W) = G_1 W$, *and so* $G_1 W = \widetilde{W} G_1^*$.

Moreover, if we write $(P_n)_n$ for a sequence of orthogonal polynomials with respect to W, then they are eigenfunctions of the second order difference operator

$$\Delta\nabla(\cdot) G_2 + \Delta(\cdot) G_1, \tag{5.6}$$

and the sequence $(\nabla P_{n+1})_{n \geq 0}$ is orthogonal with respect to $\widetilde{W}$.

Proof. To simplify the writing, we set

$$\widetilde{J} = \frac{N-1}{a|v_{N-1}|^2} I + J, \tag{5.7}$$

so that

$$\widetilde{W} = \sum_{x=1}^{\infty} \frac{a^x}{(x-1)!} (I+A)^x \widetilde{J} (I+A^*)^x \delta_x.$$

We assume the following claim which will be proved later.

Claim. For a nonnegative integer x, we have

$$G_2(x)(I+A)^x = (I+A)^x \widetilde{J} x, \tag{5.8}$$

$$G_1(x)(I+A)^x = (I+A)^x \left[a(I+A)\widetilde{J}(I+A^*) - \widetilde{J}x\right]. \tag{5.9}$$

We now proceed in three steps.

Step 1. $\widetilde{W} = G_2 W$ and $G_2 W = W G_2^*$.

Indeed, since $G_2(0) = 0$, we have $\widetilde{W}(0) = 0 = G_2(0)W(0)$. For $x \geq 1$, using (5.8), we get

$$\begin{aligned}\widetilde{W}(x) &= \frac{a^x}{(x-1)!}(I+A)^x \widetilde{J}(I+A^*)^x = \frac{a^x}{x!}(I+A)^x \widetilde{J}x(I+A^*)^x \\ &= \frac{a^x}{x!}G_2(x)(I+A)^x(I+A^*)^x = G_2(x)W(x).\end{aligned}$$

Since $\widetilde{W}$ is Hermitian, we also get the Hermitian condition $G_2 W = W G_2^*$.

Step 2. $\Delta(G_2 W) = G_1 W$ and $G_1 W = W G_1^*$.

Since $G_2(0) = 0$, we have (see (2.4))

$$\Delta(G_2 W) = \sum_{x=0}^{\infty} (G_2(x+1)W(x+1) - G_2(x)W(x))\delta_x.$$

Using now (5.9) and the first step, we have for $x \geq 0$,

$$\begin{aligned}\Delta(G_2 W)(x) &= \Delta(\widetilde{W})(x) = \widetilde{W}(x+1) - \widetilde{W}(x) \\ &= \frac{a^x}{x!}(I+A)^x \left[a(I+A)\widetilde{J}(I+A^*) - \widetilde{J}x\right](I+A^*)^x \\ &= \frac{a^x}{x!}G_1(x)(I+A)^x(I+A^*)^x = G_1(x)W(x).\end{aligned}$$

Then $G_1 W = \Delta(G_2 W) = \Delta(W G_2^*) = W G_1^*$.

Step 3. The polynomials $(P_n)_n$ are eigenfunctions of the second order difference operator (5.6) and the polynomials $(\nabla P_{n+1})_{n\geq 0}$ are orthogonal with respect to the weight matrix $\widetilde{W}$.

The first statement follows straightforwardly from Theorem 2.

The definition of G_2 and G_1 (see ((5.4) and (5.5)) show that

$$(n-1)G_{22} + G_{11} = n(I+A)^{-1} + J - \left(n - 1 + N + \frac{N-1}{a|v_{N-1}|^2}\right) I - aA.$$

Since this matrix is upper triangular, we get

$$\det((n-1)G_{22} + G_{11}) = \prod_{i=1}^{N} \left(1 - i - \frac{N-1}{a|v_{N-1}|^2}\right) \neq 0$$

(since $a > 0$). The orthogonality of $(\nabla P_{n+1})_{n\geq 0}$ with respect to the weight matrix $\widetilde{W}$ now follows from Theorem 1 and steps 1 and 2.

We now prove the claim (in two steps).

Step A. For a nonnegative integer x, we have

$$G_2(x)(I+A)^x = (I+A)^x \widetilde{J} x.$$

Taking into account the definition of G_2 (5.4), this is equivalent to

$$(I+A)^x \widetilde{J} - \widetilde{J}(I+A)^x = -Ax(I+A)^{x-1}. \tag{5.10}$$

We now call $F_1(x) = (I+A)^x \widetilde{J} - \widetilde{J}(I+A)^x$ and $F_2(x) = -Ax(I+A)^{x-1}$. Since A is a nilpotent matrix of order N, F_1 and F_2 are both polynomials of degree not bigger than N. So, it will be enough to prove that

$$\Delta^k F_1(0) = \Delta^k F_2(0), \quad k = 0, \dots, N.$$

Since

$$\Delta^k (I+A)^x = (I+A)^x A^k = A^k (I+A)^x, \tag{5.11}$$

$$\Delta^k \left(x(I+A)^x\right) = x(I+A)^x A^k + k(I+A)^{x+1} A^{k-1}, \tag{5.12}$$

a simple computation gives

$$\Delta^k F_1(x) = A^k (I+A)^x \widetilde{J} - \widetilde{J}(I+A)^x A^k,$$
$$\Delta^k F_2(x) = -xA^{k+1}(I+A)^{x-1} - kA^k (I+A)^x.$$

And hence for $k = 0, \dots, N$.

$$\Delta^k F_1(0) = A^k \widetilde{J} - \widetilde{J} A^k, \qquad \Delta^k F_2(0) = -kA^k.$$

A simple computation using the definition of $\widetilde{J}$ (5.7) gives

$$A^k \widetilde{J} - \widetilde{J} A^k = -k \sum_{i=1}^{N-1} v_i v_{i+1} \cdots v_{i+k-1} \mathcal{E}_{i,i+k} = -kA^k.$$

And the step A is proved.

Step B. For a nonnegative integer x, we have

$$G_1(x)(I+A)^x = (I+A)^x \left[a(I+A)\widetilde{J}(I+A^*) - \widetilde{J}x\right].$$

The constraints (5.3) on the parameters v_i are used in this step (and hence in the step 2 above). We proceed as before. Call

$$H_1(x) = (I+A)^x \left[a(I+A)\widetilde{J}(I+A^*) - \widetilde{J}x\right],$$
$$H_2(x) = G_1(x)(I+A)^x.$$

Again, H_1 and H_2 are both polynomials of degree not bigger than N. Hence it will be enough to prove that for $k = 0, \dots, N$

$$\Delta^k H_1(0) = \Delta^k H_2(0).$$

Writing $G_1(x) = G_{11}x + G_{10}$, a simple computation gives for $k = 0$

$$H_1(0) = a(I+A)\widetilde{J}(I+A^*) = G_{10} = H_2(0).$$

For $k \geq 1$, from (5.11) and (5.12) we get straightforwardly

$$\Delta^k H_1(x) = (I+A)^x \left[a(I+A)A^k \widetilde{J}(I+A^*) - xA^k \widetilde{J} - kA^{k-1}(I+A)\widetilde{J}\right],$$
$$\Delta^k H_2(x) = \left[xG_{11}A + kG_{11}(I+A) + G_{10}A\right](I+A)^x A^{k-1}.$$

Hence,

$$\Delta^k H_1(0) = (I+A)A^{k-1}\left[aA\widetilde{J}(I+A^*) - k\widetilde{J}\right],$$
$$\Delta^k H_2(0) = k\left[G_{11}(I+A) + G_{10}A\right]A^{k-1}.$$

It is easy to see that the equality $\Delta^k H_1(0) = \Delta^k H_2(0)$ can be rewritten in the form

$$A^k G_{10} - G_{10} A^k = k G_{11}(I + A)A^{k-1} + k(I + A)A^{k-1}\widetilde{J}.$$

The only component of this equation appears in the positions $(i, i + k - 1)$, $(i, i + k)$ and $(i, i + k + 1)$ of the matrix, and a careful computation shows that the equation holds by using (1.7). The step B is also proved. □

Write $\mathcal{A}$ for the set formed by all nilpotent matrices $A = A(v_1, \ldots, v_{N-1})$ defined by (1.5) whose parameters $v_1, \ldots, v_{N-1}$ satisfy the constraints (5.3). For a matrix $A \in \mathcal{A}$, we denote by W_A the weight matrix (5.1). We now prove that, for every matrix $A \in \mathcal{A}$ the weight $\widetilde{W}$ (5.2) is, up to a normalization and a shift in the variable, again of the form $W_{\widetilde{A}}$ for a certain matrix $\widetilde{A} \in \mathcal{A}$.

Proposition 5.1. *Given non null complex numbers v_i, $i = 1, \ldots, N-1$, satisfying the constraints (5.3), define new complex numbers w_j, $i = j, \ldots, N-1$, by*

$$w_j = \frac{\lambda_{j+1}}{\lambda_j} v_j, \tag{5.13}$$

where $\lambda_j = \sqrt{(N-j)a|v_{N-1}|^2 + N - 1} \in \mathbb{R}$. Write $A = A(v_1, \ldots, v_{N-1})$ and $\widetilde{A} = A(w_1, \ldots, w_{N-1})$. Then

1. *The parameters w_i, $i = 1, \ldots, N-1$, also satisfy the constraints (5.3), and hence $\widetilde{A} \in \mathcal{A}$.*
2. *Moreover, if we write M for the matrix of numbers*

$$M = \frac{1}{v_{N-1}}(I + A)\begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_N \end{pmatrix}, \tag{5.14}$$

then, $\widetilde{W}(x) = M W_{\widetilde{A}}(x-1) M^*$, $x \geq 1$, where $\widetilde{W}$ is the weight matrix (5.2).

Proof. Part 1 is an easy consequence of the definition of w_j and (5.3).

In order to prove part 2, we have to check that for $x \geq 1$

$$\frac{a^x}{(x-1)!}(I + A)^x \widetilde{J}(I + A^*)^x = \frac{a^{x-1}}{(x-1)!} M(I + \widetilde{A})^{x-1}(I + \widetilde{A}^*)^{x-1} M^*,$$

where $\widetilde{J}$ is defined by (5.7).

The previous identity is a consequence of the following one

$$M(I + \widetilde{A})^{x-1} = \sqrt{a}(I + A)^x \widetilde{J}^{1/2}.$$

We denote then $E_1(x) = M(I + \widetilde{A})^{x-1}$ and $E_2(x) = \sqrt{a}(I + A)^x \widetilde{J}^{1/2}$, and proceeding as in the proof of Theorem 3 (steps A and B), we prove that $\Delta^k E_1(1) = \Delta^k E_2(1)$ for $k = 0, \ldots, N-1$. Since both E_1 and E_2 are polynomials of degree $N-1$, we then have $E_1 = E_2$. Using that $\Delta^k (I + A)^x = (I + A)^x A^k$, we get

$$\Delta^k E_1(1) = M\widetilde{A}^k, \qquad \Delta^k E_2(1) = \sqrt{a}(I + A)A^k \widetilde{J}^{1/2}.$$

We have then to prove that

$$M\widetilde{A}^k = \sqrt{a}(I+A)A^k\widetilde{J}^{1/2}.$$

But this identity follows from the definition of M if we take into account that

$$\sqrt{a}\widetilde{J}^{1/2} = \frac{1}{v_{N-1}}\begin{pmatrix}\lambda_1 & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & \cdots & \lambda_N\end{pmatrix},$$

$$\widetilde{A} = \begin{pmatrix}\lambda_1 & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & \cdots & \lambda_N\end{pmatrix}^{-1} A \begin{pmatrix}\lambda_1 & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & \cdots & \lambda_N\end{pmatrix}. \quad \square$$

Corollary 5.2. *Let W be the weight matrix defined by (5.1), where we assume that the parameters v_i, $i = 1,\dots,N-2$, satisfy the constraints (5.3). Then for $k \geq 1$, the polynomials $(\nabla^k P_{n+k})_n$ are again orthogonal with respect to a weight matrix.*

The weight matrices (5.1) were introduced in [1] without the constraints (5.3). It is proved there that the orthogonal polynomials with respect to W are eigenfunctions of the following second order difference operator

$$\widetilde{D} = \Delta\nabla(\cdot)(I+A)^{-1}x + \Delta(\cdot)\left[a(I+A) - (I+A)^{-1}x\right] + (\cdot)\left[a(I+A) - J\right].$$

According to Theorem 1, when the parameters v_i, $i = 1,2,\dots,N-1$, satisfy the constraints (1.7), the orthogonal polynomials $(\nabla P_n)_{n\geq 1}$ are also eigenfunctions of the second order difference operator (5.6). This difference operator is however linearly independent to the one found in [1]. This phenomenon never happens in the scalar case where the elements of each of the classical discrete families of orthogonal polynomials are eigenfunctions of only one linearly independent second order difference operator, respectively (see [5] for more details).

Among other things, we next show that for the weight matrix W (5.1) the constraints (5.3) seem to be necessary for the existence of polynomials G_2 and G_1 (of degrees at most 2 and 1, respectively) satisfying the Pearson equation $\Delta(WG_2^*) = WG_1^*$ and the Hermitian condition $G_2W = WG_2^*$.

Example 2. Consider the weight matrix

$$W = \sum_{x=0}^{\infty} \frac{a^x}{x!}(I+A)^x(I+A^*)^x\delta_x, \tag{5.15}$$

where A is now an arbitrary upper triangular nilpotent matrix of size 3×3

$$A = \begin{pmatrix}0 & v_1 & v_3\\ 0 & 0 & v_2\\ 0 & 0 & 0\end{pmatrix},$$

where $v_1, v_2 \neq 0$.

Using Maple, it is straightforward to show that there always exist polynomials G_2 and G_1 of degrees not bigger than 2 and 1, respectively, such that $\Delta(WG_2^*) = WG_1^*$ (only linear equations

are needed). In fact, the dimension of the linear subspace

$$\{G_2 \in \mathbb{P}_2^N : \text{there exists } G_1 \in \mathbb{P}_1^N \text{ with } \Delta(WG_2^*) = WG_1^*\}$$

is in general equal to 9.

However, it is not difficult to see that only when $v_3 = 0$ and the parameters v_1 and v_2 satisfy (5.3), one of these polynomials satisfies also the Hermitian condition $G_2W = WG_2^*$. This means that the polynomials $(\nabla P_{n+1})_{n\geq 0}$ are always orthogonal with respect to a moment functional but this moment functional can be represented by a weight matrix (in the sense defined in the introduction) only when $v_3 = 0$ and the parameters v_1 and v_2 satisfy the constraints (5.3).

Take now particular non null values of the parameters. Using Maple it is not difficult to generate the first few orthogonal polynomials with respect to W. For instance for $v_1 = v_2 = v_3 = 2$ and $a = 1$, we have

$$P_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$P_1 = \begin{pmatrix} x - \frac{329}{69} & \frac{314}{69} & -\frac{2}{69} \\ -\frac{58}{69} & x - \frac{17}{69} & -\frac{10}{69} \\ -\frac{2}{69} & -\frac{22}{69} & x - \frac{17}{69} \end{pmatrix}$$

$$P_2 = \begin{pmatrix} x^2 - \frac{5051}{569}x + \frac{5017}{569} & \frac{7788}{569}x - \frac{3308}{569} & -\frac{11300}{569}x + \frac{136}{569} \\ -\frac{500}{569}x + \frac{1068}{569} & x^2 - \frac{1411}{569}x - \frac{137}{569} & \frac{404}{569}x + \frac{4}{569} \\ -\frac{4}{569}x + \frac{136}{569} & -\frac{380}{569}x + \frac{124}{569} & x^2 - \frac{411}{569}x + \frac{41}{569} \end{pmatrix}.$$

One can then check that the polynomials P_n, $n \geq 0$, cannot be common eigenfunctions of a second order difference operator of the form (1.6) (only linear equations are needed). This shows that the hypothesis $G_2W = WG_2^*$ cannot be removed in Theorem 2.

Example 3. We finally display a weight matrix W whose orthogonal polynomials $(P_n)_n$ are eigenfunctions of a second order difference operator, however this weight matrix W does not satisfy any discrete Pearson equation. Indeed, consider the discrete 2×2 weight matrix defined by

$$W = \sum_{x=0}^{k} \frac{\Gamma(k+1)}{x!\Gamma(k+1-x)} \begin{pmatrix} 1 & \frac{v_1}{k+1}x(x-k) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{v_1}{k+1}x(x-k) \\ 0 & 1 \end{pmatrix}^* \delta_x$$

where k is a positive integer. This weight matrix is a particular case of an $N \times N$ weight matrix introduced in [1]. The orthogonal polynomials $(P_n)_{n=0}^{k-1}$ with respect to W (they are now finitely many) are also eigenfunctions of a second order difference operator. Using Maple is easy to check that there do not exist any polynomials G_2 and G_1 of degrees not bigger than 2 and 1, respectively, such that $\Delta(WG_2^*) = WG_1^*$ (only linear equations are needed). In particular, the family $(\nabla P_{n+1})_{n\geq 0}$ cannot be orthogonal with respect to any moment functional. This shows that in the matrix orthogonality, the discrete Pearson equation for the weight matrix W is independent

of whether the orthogonal polynomials with respect to W are eigenfunctions of a second order difference operator of the form (1.6).

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